Bilby for individual gravitational wave transients and beyond

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Bilby

Bilby: a user-friendly Bayesian inference library.

The aim of bilby is to provide a user-friendly interface to perform parameter estimation. It is primarily designed and built for inference of compact binary coalescence events in interferometric data, but it can also be used for more general problems.

The code, and many examples are hosted at https://github.com/bilby-dev/bilby. For installation instructions see https://bilby-dev.github.io/bilby/installation.html.

What is Bilby used for

Gravitational waves (not exhaustive)

- Compact binaries
  - Since GWTC-3 bilby_pipe has been the primary engine for Bayesian inference in the LVK (including many TGR and lensing analyses)
  - Other inference codes, e.g., DINGO uses Bilby for many GW operations
  - GWPopulation has produced headline population results for LVK papers since O3.
  - LVK cosmological inference codes also use Bilby
- Continuous waves (cwinpy)
- Stochastic (pygwb)
- Burst
- Detchar (Bayesian glitch fitting)
- …

What is Bilby used for

Not ground based gravitational waves (not exhaustive, mostly taken from GitHub)

- LISA
- Multimessenger astronomy
  - NMMA, redback, ...
- FRBs

- Other things I've seen
  - Neuroscience
  - Image recognition

2 new contributors in last 90 days:
Filippo Santoliquido, Meesum Qazalbash

8 total contributors in last 90 days

last 90 days: 6 issues closed, 7 opened
29 PRs closed, 34 opened

papers citing bilby: 40 last month
154 this year
1295 all-time

Generated on April 21, 2026

Inference the Bilby way

- Bilby is a library not a monolith
- Use it for scripting as you would any other library, e.g., numpy
- If you have a specific use case, see if an implementation using Bilby already exists.
- If you find a bug, open an issue, or a pull request!
- Try not to directly modify Bilby, if you think some functionality can be improved open an issue
- If you have a new sampling algorithm, consider adding a Bilby plugin

Pause

bilby_pipe

a.k.a., production PE with Bilby

- Bilby provides a huge amount of flexibility
- Reliable and reproducible deployment on high-throughput/high-performance computing requires requires pipelining code.
- bilby_pipe provides a standardized method for analyzing gravitational-wave transients that understands LVK data acquisition and handling
- Used by LVK for online PE and production PE from O3-present

bilby_pipe

a.k.a., production PE with Bilby

- Interface via a configuration file with *many* options (we'll work through some later)
- Handles automatically
  - Data finding
  - Submission to HTCondor or SLURM
  - Distribution over the OSG using HTCondor

Pause

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- Bilby is fully open source and contributions are welcome from anyone, especially bug and documentation fixes.
- Feature requests are considered case-by-case, however, we often recommend initially packaging up new functionality as a package that uses Bilby.
- For LVK work we are strongly person-power limited. If you’re interested on working on the next iteration of bilby_pipe come talk to me!